EXACTNESS OF THE CECH SEQUENCE OF THE RING ITSELF IS EQUIVALENT TO BIJECTIVITY ON F-POWER TORSION, GIVING A GLUEING PAIR CRITERION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 25254-25270 (canonical tag 0BNR) and the counted proof is the proof environment at source lines 25272-25298, which immediately follows it and is its own proof.

Editorial note. The statement refers to the displayed Cech sequence by its label equation-BL-cech-re, which lies outside the counted spans; that label is carried as a navigation stub so the reference resolves in the compiled document.

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Allowed standard prerequisites (editorial).

- Every element of the localization of the target ring at the element can be written with a denominator, and the difference between a lift and the numerator can be absorbed by a multiple of a power of the element using that the quotients by the powers of the element agree. used at proof lines 25273-25276. Gives exactness on the right by producing an explicit pair mapping to a given element of the localization.
- The kernel of the localization map of a ring is the ideal of elements annihilated by a power of the element. used at proof line 25279. Gives the left exactness criterion.
- The last statement of the lemma is a special case of the penultimate one, with the completion of the ring in the role of the second ring. used at proof lines 25296-25297. Transfers the glueing pair criterion from an arbitrary second ring to the completion.

Source reference. Original equation 0F1Q: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25238–25238. The reference points to the displayed Cech sequence that the statement is about; it names the object under discussion and the element computations deciding its exactness are inside the counted span.

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Source reference. Original lemma 0BNJ: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25129–25129. The reference is invoked at proof lines 25296-25297 to obtain the completion case as a special case of the penultimate statement; it is the final step and the rest of the proof is inside the counted span.

Lemma 0.2. *Let R be a ring, let $f \in R$, and let $R \to R'$ be a ring map which induces isomorphisms $R/f^nR \to R'/f^nR'$ for $n > 0$. The sequence (0F1Q) is*

(1) *exact on the right,*
(2) *exact on the left if and only if $R[f^\infty] \to R'[f^\infty]$ is injective, and*
(3) *exact in the middle if and only if $R[f^\infty] \to R'[f^\infty]$ is surjective.*

In particular, $(R \to R', f)$ is a glueing pair if and only if $R[f^\infty] \to R'[f^\infty]$ is bijective. For example, (R, f) is a glueing pair if and only if $R[f^\infty] \to R^\wedge[f^\infty]$ is bijective.

Proof. Let $x \in R'_f$. Write $x = x'/f^n$ with $x' \in R'$. Write $x' = x'' + f^n y$ with $x'' \in R$ and $y \in R'$. Then we see that $(y, -x''/f^n)$ maps to x. Thus (1) holds.

Part (2) follows from the fact that $\mathrm{Ker}(R \to R_f) = R[f^\infty]$.

If the sequence is exact in the middle, then elements of the form $(x, 0)$ with $x \in R'[f^\infty]$ are in the image of the first arrow. This implies that $R[f^\infty] \to R'[f^\infty]$ is surjective. Conversely, assume that $R[f^\infty] \to R'[f^\infty]$ is surjective. Let (x, y) be an element in the middle which maps to zero on the right. Write $y = y'/f^n$ for some $y' \in R$. Then we see that $f^n x - y'$ is annihilated by some power of f in R'. By assumption we can write $f^n x - y' = z$ for some $z \in R[f^\infty]$. Then $y = y''/f^n$ where $y'' = y' + z$ is in the kernel of $R \to R/f^nR$. Hence we see that y can be represented as $y'''/1$ for some $y''' \in R$. Then $x - y'''$ is in $R'[f^\infty]$. Thus $x - y''' = z' \in R[f^\infty]$. Then $(x, y'''/1) = (y''' + z', (y''' + z')/1)$ as desired.

The last statement of the lemma is a special case of the penultimate statement by Lemma 0BNJ. □